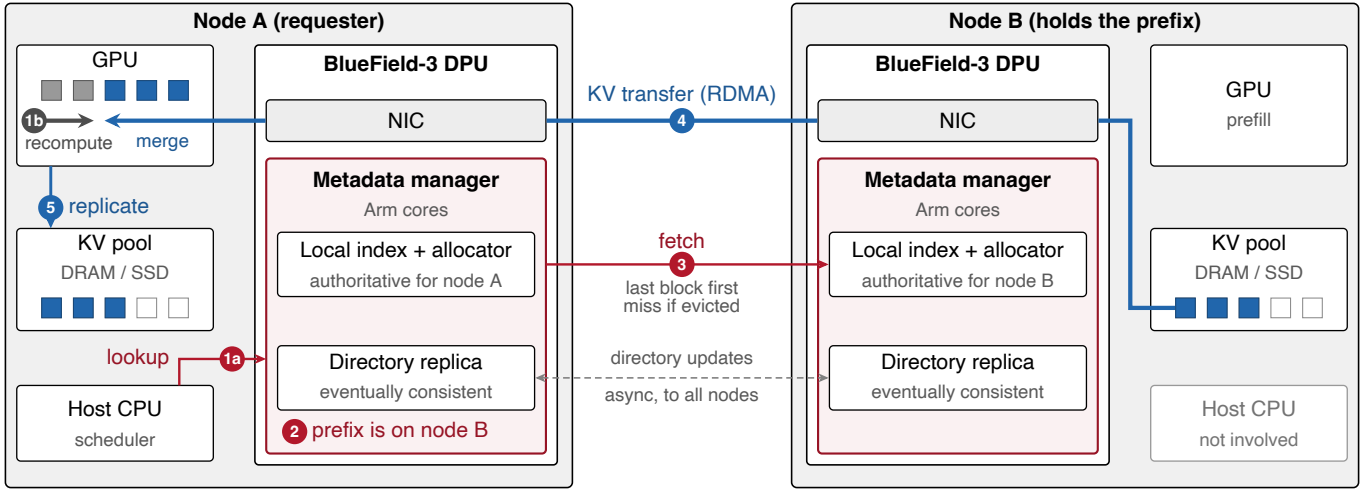
Node A (requester)
GPU
1b
recompute
merge
BlueField-3 DPU
NIC
KV transfer (RDMA)
4
Node B (holds the prefix)
BlueField-3 DPU
NIC
GPU
prefill
5
replicate
Metadata manager
Arm cores
Local index + allocator
authoritative for node A
fetch
3
last block first
miss if evicted
Metadata manager
Arm cores
Local index + allocator
authoritative for node B
KV pool
DRAM / SSD
KV pool
DRAM / SSD
lookup
1a
Directory replica
eventually consistent
directory updates
async, to all nodes
Directory replica
eventually consistent
Host CPU
scheduler
2
prefix is on node B
Host CPU
not involved